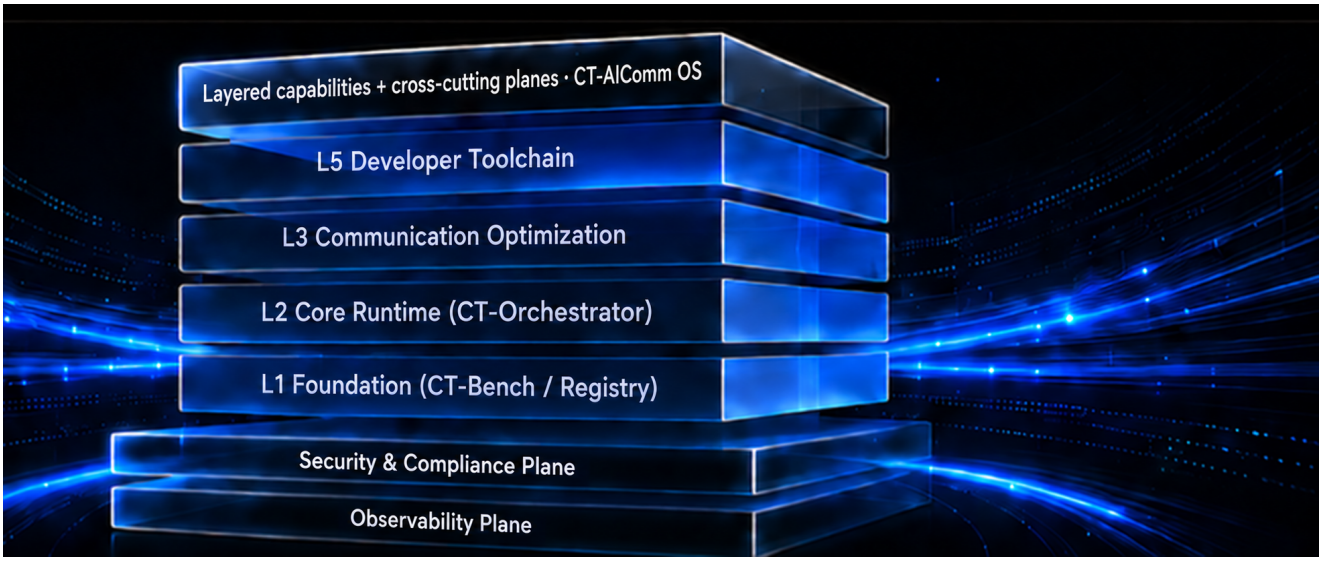
OPEN AI COMMUNICATION INFRASTRUCTURE

CT-AIComm

An operating system for AI communication decisions

Build the AI communication operating system for cross-domain coordination.



EDITION	STATUS	EVIDENCE	PRODUCTION
2026.10	Public working paper	Explicitly scoped	No autonomous control claim

A public, evidence-first system for coordinating protocols, network resources, policy constraints, measurements, and named human approvals across networks, clouds, and accelerators.

Executive summary

AI training and inference depend on heterogeneous communication paths, but protocol knowledge, resources, policy constraints, and measurement evidence are usually managed in separate systems. CT-AIComm treats this fragmentation as an operating-system problem.

The product connects a public Registry, CT-Bench, CT-DSL, CT-Orchestrator, observability, compliance evidence, and enterprise adoption. It creates one traceable workflow from discovery to a human-owned decision without turning a recommendation into execution authority.

Core workflow

- Discover versioned communication resources in the Protocol Registry.
- Declare latency, cost, region, encryption, and failure constraints in CT-DSL.
- Prepare explainable, fail-static recommendations through CT-Orchestrator.
- Validate claims with environment fingerprints, raw samples, methods, failures, and checksums.
- Retain named human approval for publication, deployment, procurement, compliance, and certification.

Evidence boundaries

A catalog record is not a benchmark result. A synthetic simulation is not a measured run. A recommendation is not execution authorization. A readiness artifact is not external certification.

Why the operating-system model matters

Communication decisions cross protocol, infrastructure, policy, cost, and governance boundaries. Treating each concern as a separate tool produces incompatible inventories and weak handoffs. CT-AIComm preserves one versioned decision record while allowing each domain to keep its own authority.

Architecture and evidence

L1 - Foundation

The Protocol Registry stores versioned protocol resources and provenance. CT-Bench stores scenario definitions, environment fingerprints, raw observations, failures, aggregates, and checksums. Catalog evidence and benchmark evidence remain separate.

L2 - Core runtime

CT-Orchestrator evaluates registered resources against declared constraints, produces deterministic recommendations, records reasons, and fails static when no route is eligible. It does not silently change production traffic.

L3 - Communication optimization

Optimization contracts expose protocol and topology choices while retaining the workload, environment, scoring rule, and policy boundary that made a result meaningful.

L4-L5 - Governance and developer tooling

Security, compliance, policy review, release evidence, SDKs, and workflow tools make decisions inspectable. Named reviewers remain responsible for approvals and external assessments.

L0 - Observability

Telemetry, audit records, release provenance, and bounded operational evidence show what the system observed and prepared. Observability does not manufacture certainty when evidence is absent.

Minimum reproducible Bench package

- Methodology version and stable field definitions with explicit units.
- Hardware, kernel, driver, runtime, protocol, topology, and policy fingerprint.
- Dependency lock and one-command runner.
- Raw CSV, retained failures, aggregates, and checksums.
- Visible synthetic or measured origin plus independent review state.

Governance and adoption

CT-AIComm separates submitters, reviewers, administrators, operators, and external assessors. It also separates cataloged, eligible, recommended, approved, prepared, acknowledged, and independently verified states. These boundaries prevent a score or recommendation from becoming an implicit production change.

Adoption path

- Choose one bounded workload and record data-region, security, and operational constraints.
- Create an environment fingerprint and a reproducible baseline before optimization.
- Import authentic measurements with raw evidence; do not relabel synthetic examples.
- Review risks, rollback conditions, and ownership before any controlled traffic change.
- Record a named decision and retain evidence for replay and external assessment.

Current public scope

The implementation includes Registry, synthetic demonstration runs, reproducible package generation, public read APIs, deterministic policy tools, governance workflows, and enterprise evidence templates. Verified real benchmark claims remain zero until authentic results are independently reviewed.

Public resources

Product: cloudtelecom.cn/en/products/
Registry: cloudtelecom.cn/en/registry/
Bench: cloudtelecom.cn/en/bench/
Documentation: cloudtelecom.cn/en/docs/
Engineering notes: cloudtelecom.cn/en/blog/

Public working paper. This document describes product architecture and evidence boundaries. It is not a certification, procurement commitment, SLA, production authorization, or protocol performance ranking.